

Core Framework How-To Guide

Business Process Framework (eTOM)

User Guidelines for the Business Process Framework

GB921 Addendum U

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Executive Summary

This document is an Application Note attached to The Business Process Framework (GB921). It is intended to provide users of the Business Process Framework (eTOM) with guidelines and information to assist them in applying the Business Process Framework within their businesses.

21.5 Release notice

PLEASE NOTE:

Because of some major adjustments to domains in ODA and additional structural changes to the Business Process Framework, significant amendments to diagrams and text has commenced. It is expected to be completed in subsequent releases. The interim state may cause inconsistencies in this and other ODA documents. Specifically, there are known diagram depictions, term conflicts, and numbering issues that are being addressed.

1. Introduction

This document explains the design decisions of the Business Process Framework and sets out the principles for its application and extension in the form of Guidelines. These Guidelines are of use in two particular cases. First, for the practitioner or process architect who wants to apply the Business Process Framework in a consistent way to specific situations. Second, for assurance of auditing and traceability, in order to produce repeatable application of the Business Process Framework within an organization and to demonstrate this externally for audit purposes.

It also provides a basis for contributors to the Business Process Framework and the evaluation of contributions by the Business Process Framework team. Note that the concepts in this document have emerged throughout the development lifecycle of the Business Process Framework itself, and as a consequence, historical design decisions that are embedded within the Business Process Framework may deviate from the recommended practice herein that is intended to guide ongoing users of the Framework. The intent is to converge the practice here with the Business Process Framework as it is developed further, but users should recognize that this will be an evolving situation.

It is intended that the guidance in this document will be augmented through user experience and feedback. A list of outstanding issues is included.

1.1 Audience

The intended audience for the Guidelines is practitioners, process architects, and process auditors. The material here is advanced material - it is not introductory (as that information is available elsewhere in the Business Process Framework set).

1.2 Plan of document

The document is set out as follows:

- The principles and positions of the Business Process Framework model
- Process patterns
- Relationship of the Business Process Framework to ODA and the ODA elements
- Business Process Framework and an organization's own processes
- Guidance for extending and using the Business Process Framework
- Audit checklist

1.3 Principles and Guidelines

The principles are shown in the following format. Optionally, a Best Practice Guideline is indicated where a recommendation is made about the way in which the Business Process Framework should be applied. Requirements for auditing purposes are also shown in this format.

Principle Business Process Framework.nn Business Process Framework is ...
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2. Principles and positions of the Business Process Framework model

2.1 Introduction

This section sets out to position the Business Process Framework as a process framework and to make explicit the modeling and design choices which have been followed in its development.

2.2 A note on the status of the Business Process Framework documentation

The Business Process Framework consists of both normative and non-normative material. The normative material is the Standard; the non-normative material is included for information and guidance. In general, GB921, with its Annexes and Addenda is normative material; Appendices and Application Notes are non-normative. Process Descriptions in Addendum D and DX, Process Flow Examples in Addendum F and E are all normative. Other TMF documents, such as GB939 (Business Services Examples), which include Business Process Framework material, are non-normative.

There is, however, an important distinction between the Process Elements (PEs), presented in Addendum D and DX, and the Process Flows, presented in Addendum F and E. Both are normative, i.e., part of the Standard, but the Process Elements are comprehensive in scope while the Process Flows are examples.

2.3 Nature of the Business Process Framework model

The Business Process Framework is a set of Process Elements, which are organized into a hierarchical framework. The Process Elements are activity-based, and the Business Process Framework is thus an activity-based process decomposition model.

Principle Business Process Framework - 01
Business Process Framework is an activity/task-based process decomposition model.

The Process Elements in the Business Process Framework are intended as an exhaustive list; they are comprehensive in scope. It is the intent that all business activities and tasks in the Enterprise can be supported by (i.e., are able to fit within) the Business Process Framework Process Elements. Each Process Element has a detailed description that can include the purpose, inputs, outputs, interfaces, high level information and functional requirements and business rules.

Principle Business Process Framework - 02
Business Process Framework Process Elements are comprehensive for a Service Provider.

Note:

The Business Process Framework is a process model, not a state model. It contains processes, not states. For example, it contains process definition and decomposition for the processes of Order Handling but does not model the different states of an Order.

2.4 Process model types

In general, there are 3 approaches to the modeling of business processes. It is also possible to use a hybrid of these approaches. The general approaches are:

- activity-based process modeling
- communication-based process modeling
- artifact-based process modeling

Activity-based process modelling

Here the overall process is decomposed into tasks that are ordered based on the dependencies among them. The fundamental entity of a business process for the Activity-based approach is the unit of work and a business process is considered to be a succession of activities, or units of work, following a specific control flow.

Definition: Activity

An activity represents a unit of work performed by a party or system. Activities transform inputs into outputs and are associated with triggers and outcomes (pre and post conditions).

Principle Business Process Framework - 03
A Business Process Framework Process Element is a succession of activities with a control flow.

Communication-based Process Modelling

In this approach, an action in a process flow is represented by the communication that flows from a provider toward a consumer (In process improvement, a SIPOC is a tool that summarizes the inputs and outputs of one or more processes in terms of: Supplier=>Input=>Process=>Output=>Consumer). In the communication-based approach the communication is the message. So, a business process can be expressed as an exchange of messages, or transaction, between two or more roles and every state change within a company can be associated with the processing of a message.

Artifact-based Process Modelling

In the artifact-based approach objects, or artifacts, are created, modified and used during the process and thus the model is based on work products and their paths through a series of workflow activities.

Hybrid approach to Process Modelling

The hybrid approach uses a combination of these general approaches to produce a set of models for an organization's processes. Typical models might be based on an information flow model (from the communication-based approach), a capabilities model (from the artifact-based approach) and a process-model (activity-based approach).

2.5 Characteristics of a Business Process

In general, a Business Process will have the following characteristics:

- It has a goal
- It has specific inputs
- It has specific outputs
- It transforms inputs into outputs
- It uses resources
- It has several activities that are performed in some order
- It creates value of some kind for the customer. The customer may be internal or external.

In addition:

- It may affect more than one organizational unit - "Horizontal organizational impact"
- Its effects on information entities can be analyzed via CRUD (Create / Read / Update / Delete)
- It may have a responsibility model for the roles associated with the process, expressible as RACI Characteristics (Responsible / Accountable / Consulted / Informed)

For a Business Process Framework Process Element:

- Goal is stated.
- Inputs may be defined.
- Outputs may be defined.
- Resources consumed may be defined.
- Activities may be specified within description
- Value should be stated.
- Effect on organizational use may be stated
- CRUD analysis may be available

- RACI analysis may be available

Principle Business Process Framework - 04

(Best Practice): a Business Process Framework Process Element has a goal, an implied purpose or value proposition, inputs, outputs. It consists of activities/tasks and uses resources. It may have a CRUD and a RACI model.

2.6 Decomposition

Definition: Decomposition

Decomposition is the breaking-down of a process into simpler activities.

The Business Process Framework is a decomposition model from a notional Level 0 through to Level 4, and beyond (some Level 5 process elements are now defined and are being incorporated). Additionally, many individual users are developing lower-level decompositions that extend the Business Process Framework beyond the industry-agreed level (and in due course these may feedback to extend the level of industry agreement). In order to keep the Business Process Framework to a level which is generally useful it is not intended to decompose the Business Process Framework (i.e., as managed through the TM Forum) indefinitely. It is asserted that the further a decomposition is taken, the more difficult it is to prove the uniqueness of lower level processes. However, the level at which it becomes unproductive to extend further is not yet clear, and so decomposition continues to proceed in line with the industry's priorities and available effort.

Note:

The relationship and mapping of these Business Process Framework Levels to an organization's own processes and procedures is addressed in a later section "Organizational Context for the Business Process Framework".

Principle Business Process Framework - 05

The Business Process Framework is decomposed from notional Level 0 to more granular levels – Levels 1, 2, 3 and 4 (and some of levels 5 and 6) are addressed so far. An agreed endpoint (i.e., a level below which decomposition does not proceed) is not yet defined.

Principle Business Process Framework - 06

It is not the purpose of the Business Process Framework to address the detailed processes and procedures of an enterprise.

Principle Business Process Framework - 07

The Enterprise domain is generally decomposed to Level 3 only (but specific areas that represent particular priorities have been decomposed further into level 4 and below).

2.7 Traceability

Because the Business Process Framework is a decomposition model, the lower levels of the decomposition can be traced back to the higher levels.

Principle Business Process Framework - 08

The goals, inputs, outputs, and activities of decomposed Process Elements at a lower level are consistent with the higher-level Process Element. In particular, the input of the first lower level Process Element is the same as the input of the higher level Process Element; the output from the last lower level Process Element is the same as the output of the higher level Process Element; the detailed goals of the lower level Process Elements taken together should match the goal of the higher level Process Element; the activities of the lower level Process Elements taken together should match the activity of the higher level Process Element.

2.8 Process dependency through information

Business processes do not exist in isolation. Processes require information from other processes, and they in turn provide information to other processes. Dependencies (or associations) between processes occur when an activity requires information from another activity. Process dependencies are related to the entities and attributes required by the business area. The importance of analyzing and modeling dependencies is to provide further understanding of the interaction between processes and data.

Note:

An exercise has been underway for several years to map the Information Framework (SID) ABEs (Aggregate Business Entities) which are associated with Business Process Framework Process Elements.

Principle Business Process Framework - 09

(Best Practice): each Business Process Framework Process Element should identify its associated Information Framework ABEs.

2.9 Grouping / organization within the Business Process Framework

The Business Process Framework is a classification or taxonomy of Process Elements. At Level 0 the elements are classified into S2R (Strategy to Operations) and Operations. Lower Levels are formed by decomposition with each Process Element occurring once only.

Principle Business Process Framework - 10

A particular Process Element will occur only once in the Business Process Framework; there is no replication.

2.10 Flows

Amongst the various flows that are characteristic in most organizations, there are three that are fundamental in the context of the Business Process Framework, namely the information flow, the material flow and the control flow.

- The information flow concerns the flow of data or information, e.g., the information on an order as it is progressed; these can also be message flows.
- The material flow concerns the actual physical items e.g., the items which constitute the order.
- The control flow (or workflow) defines the logic of business processes i.e., the enterprise behavior in terms of a sequence or order in which enterprise activities must be performed to achieve business objectives.

The definition of the Business Process Framework Process Elements themselves does not address these types of flow. However, the Business Process Framework does include a couple of Addenda: E and F; sample process flows and depictions of process interaction in swim lanes. These are examples of control flow.

Principle Business Process Framework - 11
Process flow examples in Business Process Framework are control flows, defining the sequence in which activities are performed.

Note:

Traceability also applies to swim lanes in Business Process Framework process flows. (See Principle Business Process Framework - 08)

Principle Business Process Framework - 12
The swim lanes in a process flow are consistent within themselves and with respect to lower level decompositions.

2.11 Dynamic aspects of Process Modeling

The Business Process Framework Process Elements and example process flows are a process view of the enterprise behavior, based on sequences of activity. However, there are also dynamic aspects pertaining to the processes and their interaction. These are considered below.

- Temporal aspects

There may be time-based requirements in the triggering of processes, triggering frequencies and possible delays between process steps. Process step durations (minimum, maximum, average durations) can also be indicated

➤ Co-operative activities

In practice, it is common that two or more activities of two different processes must work co-operatively, e.g., to exchange messages or objects. Methods include message passing and patterns.

➤ Process communication

In the case where processes must communicate, this means that some activities of one process must interact with activities of other processes. The previous mechanisms for co-operative activities can be used.

➤ Process synchronization

Process synchronization can happen in three different forms: (1) synchronization by events, (2) synchronization by messages and (3) synchronization by object flows.

➤ Exception handling mechanisms

Process models often only model the ideal structure of a business process. Real-world situations mostly consist of dealing with exceptions. Exceptions can either be predictable or unpredictable.

Principle Business Process Framework.13

Business Process Framework models success scenarios (sunny days). Error conditions are not in scope.

Principle Business Process Framework.14

Dynamic aspects of process modelling (e.g., workflow diagramming and sequential flows) are outside the scope of the Business Process Framework.

2.12 Naming conventions

The preferred convention for naming Level 3 Process Elements is <Verb Noun> e.g., “Configure & Activate Resource”, “Determine pre-order feasibility”, “Close Problem”. This is also the preferred convention for naming Process Elements at lower levels of decomposition (i.e., Level 4 and below)

The preferred convention for naming events is <Noun Verb> e.g., “Work Orders Executed”, “Resource Allocation & Configuration Done”.

Note: Level 1 and Level 2 Process Names have not in the past used the convention above. Renaming existing Level 1 and Level 2 process elements is seen as unnecessarily disruptive and so the existing form (typically, <Noun> e.g., “Party Relationship Development & Retirement”, “Order Handling”) has been retained. New or modified process elements at Level 1 and/or Level 2 should continue to use this existing convention, so that there is consistency in the naming within a Level

Principle Business Process Framework - 15

Terminology and naming conventions are <Noun Verb> for Level 1 & 2 Process Names, <Verb Noun> for Level 3 Process Names, <Noun Verb> for events

2.13 Layer References and Responsibilities

A layered approach to the handling of responsibilities and information is taken in the Business Process Framework. Responsibility for association / translation between layers is generally positioned at the lower layer. For example, the Customer layer in the Operations area manages Customer Problems and the Service layer in the Operations area manages the Service Problems that may be associated, but it is the responsibility of the processes in the Service domain to map between these Service Problems.

Thus, the Customer domain in the Operations area provides the Customer Problem (or some appropriate information from this) to the Service domain, which must then associate the one (or more) Service Problems that derive from this Customer Problem. Any ongoing interaction between Customer and Service layers is therefore in terms of Customer Problems (or information based on these) and not Service Problems, which are managed wholly within the Service layer.

Principle Business Process Framework - 16

Responsibility for association / translation between layers is generally positioned at the lower layer.

2.14 Data Responsibility

The process, which is managing data creation, update etc. has a prime responsibility for ensuring that the results of data which it is manipulating via the process are appropriately stored.

Principle Business Process Framework - 17

A process has prime responsibility for ensuring that the results of data manipulation are stored appropriately.

3. Process Patterns

3.1 Introduction

In this section are presented examples of decomposition patterns in the Business Process Framework. These patterns serve as templates for process modeling in the particular process area. Patterns are identified at Level 3 for:

- Problem Reports and their resolution
- Strategic view and Business Plan
- Order Lifecycle
- Product Lifecycle

Note:

All Patterns should have associated Use Cases. Use Cases will be added in a later version of the document.

3.2 Level 3 patterns

Example 1 – Problem Reports and their resolution

Applicable to ASSURANCE: Customer Problem Handling / Service Problem Management / Resource Trouble Management.

This pattern consists of 4 process steps and 2 continuous processes. The process steps are:

- “Create”. e.g., Problem Report, Trouble Report, Resource Trouble.
- “Analyze”. Diagnose root cause.
- “Fix”. Correct & Recover through recovery activities.
- “Close”. Problem resolved, close report.

With 2 processes running continuously:

- “Track & Manage recovery activities”
- “Report”

Pre-condition / Inputs to this pattern are: a reported problem or an alarm or event at resource or service level.

Post-condition / Output is: resolved problem, restoration of normal operation.

Associated Use Case:

Example 2 – Strategic view and Business Plan

Applicable to Strategy Management: Service Strategy & Planning / Resource Strategy & Planning / Party Strategy & Planning

This pattern consists of 6 processes.

- “Research”. Research & analyses, including management of research gathering.
- “Strategy”. Formulate strategy and business goals
- “Business Plans”
- “Operational support.”
- “Partnership.” (Null step for Supply Chain)
- “Commit.” Gain Enterprise commitment.

Pre-condition / Inputs to this pattern are research, forecasts

Post-condition / Outputs are committed business plans and strategy.

Associated Use Case:

Example 3 – Order Lifecycle

Customer Orders, Service Orders and Resource Orders through to Closure. “Track and Manage” and “Reporting” run continuously.

Associated Use Case:

Example 4 – Product Lifecycle

From Research and New Product Development through In Service to Retirement.

Associated Use Case:

3.3 Level 4 Patterns

Later issues of the document will show how the Pattern approach can be extended to Level 4. Note that these are guidelines, they do not prescribe or mandate Level 4.

4. Organizational Context for the Business Process Framework

4.1 Introduction

This section sets out the enterprise context for process modeling, and the ways in which the Business Process Framework is applied. This section is not a guide on how to do process modeling in an organization, rather it sets out a generic framework for the various types of process (including manual human procedures) within a typical enterprise and shows how the Business Process Framework can be related to those organizational processes.

4.2 Use of the Business Process Framework within an organization

The use of the Business Process Framework by organizations involves the extension and refinement of the Business Process Framework to meet the specific business, operational, system and deployment needs of the organization.

This section is based on the following view of how processes are developed and modeled within organizations, and the relationship of the process models to organization structures and systems developments.

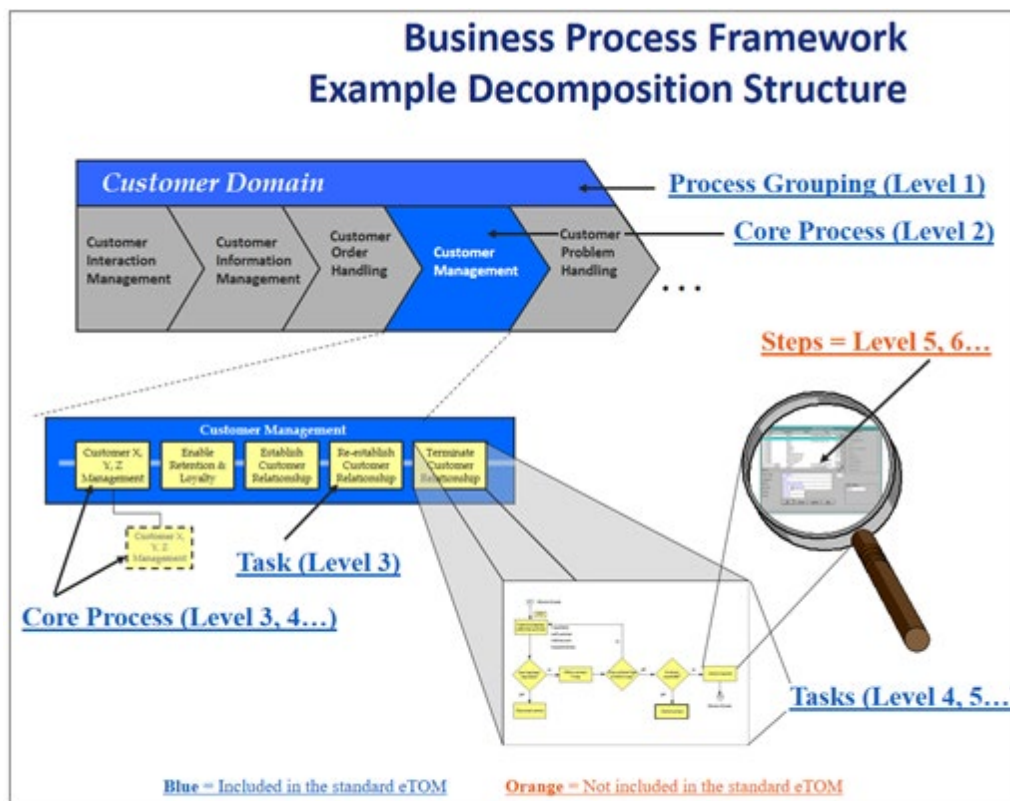


Figure 4-1 - Process Hierarchy

PLEASE NOTE: As of release 14.5 of Framework it is acceptable for a core process to decompose to other core process. Thus, multi-level core processes are being added to the Business Process Framework.

The best way to structure a large amount of content and detail, while still allowing the higher-level views to present a summary view, is to structure the information in multiple Levels, where each Level is decomposed into greater detail at the next lower Level. This is Hierarchical Decomposition.

By having the Business Process Framework structured into multiple Levels it enables users of the framework to align their enterprise framework or their process implementations with the Business Process Framework at different levels e.g., Align at Level 1 and 2 or align at Level 1, 2 and 3.

To summarize how levels are used in the Business Process Framework.

- Level 0** The whole Enterprise view (The entire Business Process Framework) as seen with its Process Areas and Domains altogether
- Level 1** Domains (horizontal) represent a Level 1 view along with the seven Context Verticals (previously referred to as Vertical Process Groupings).
- Level 2** All the Process Elements (aka Core Processes, e.g., Order Handling) which populate horizontal domains across the eTOM are Level 2.

- Level 3** Process Elements which decompose from Level 2 core processes are Level-3; these can be Core Processes themselves or Tasks.
- Level 4** Process Elements which decompose from level-3 process elements are level-4; these can also be Core Processes themselves, or Tasks.
- Level 5** Process Elements which decompose from level-4 process elements are Level 5; these could also be Core Processes (exceptions) or Tasks.
- Level 6** Process elements are rarely decomposed in the eTOM into level 6 elements, but a few examples of these exist in the Framework.
- Level 7** Process elements are rarely decomposed in the eTOM into level 7 elements, but a few examples of these may exist in the Framework.

4.3 The Business Process Framework and the Process Hierarchy

The TM Forum Business Process Framework in its analysis has addressed the concerns shown in Level 0 through to Level 4, and now it is moving to address Level 5 and beyond in selected areas.

However, the concerns for documenting lower levels in the Business Process Framework need to be addressed by an enterprise itself in implementing concrete detailed processes, roles and transactions.

For such extensions to the Framework by an enterprise, the following section provides guidance on how an enterprise should execute these analysis steps. The benefit of these guidelines is that different enterprises will use a similar analysis approach to apply the Business Process Framework to their own organization.

5. Guidance for extending and using the Business Process Framework

Because of the changes in domains and additional structural changes to the Business Process Framework, significant amendments to diagrams and text has commenced

5.1 Process Hierarchy: Decomposition Principles

Traditionally Level 0 elements in the Business Process Framework (eTOM) have been known as PAs (Process Areas) there were three PAs in the eTOM until version 14.5: SIP (Strategy, Infrastructure & Product), Ops (Operations) and EM (Enterprise Management). Process Areas as such, are not (directly) aligned with business objectives, value streams or balance scorecards; Process Areas represent a partitioning or highest-level classification of processes of the enterprise.

Currently there are two Process Areas left in the eTOM, these are: S2R (Strategy to Readiness) and Operations. Since version 14.5 the EM (Enterprise Management) Process Area has become the seventh horizontal domain in the framework and has been renamed “Enterprise” instead of “Enterprise Management”; the ‘Common’ transversal domain has been removed from the framework and its constituents refactored and distributed across various horizontal domains as appropriate.

In the past, Process Groupings represented by the 8 former horizontal domains and the 7 verticals (known also as lifecycles in earlier times) were a way of decomposing and structuring (Level 0) Process Areas into the next level of detail (Level 1s for the horizontal process groupings), these are currently seven single horizontal (Level-1) process groupings and one (Level-1) vertical; these are all better known as domains: 1) Market & Sales, 2) Customer, 3) Product, 4) Service, 5) Resource, 6) Business Partner, 7) Enterprise.

Core processes can be multi-level, typically they are Level 2 but there can be Level 3 core processes (although less common) and even Level 4 core processes. Core processes break down into Tasks.

Level 3s, Level 4s, Level 5s, Level 6s, etc. are typically known as Tasks in the eTOM, although some process models e.g., APQC define a different way of decomposition in which Level 1 (Categories) decompose into Level 2 (process groupings) which decompose into Level 3 (processes) which themselves decompose into Level 4 (activities) which decompose into Level 5 (tasks), but that is a different perspective. In fact, there are hundreds of different process models across all industry sectors and enterprises, some are standard, some are proprietary, and their structure can vary significantly, but the Business Process Framework (eTOM) provides a good balance in terms of structure and decomposition, due to its level of maturity, (telecoms) industry adoption and popularity, it is now being positioned more and more to be relevant to the full spectrum of the service provider industry, well beyond telecoms.

6. Decomposition Guidelines

Before providing a set of decomposition guidelines the properties of a process are provided. Each framework process may have one or more of these properties. As enterprise specific extensions are made to the framework, these properties should become part of the overall specification of the added processes.

A process specification consists of:

- A stated goal that the process achieves
in terms of a purpose or result from applying the process
- Inputs that may be defined
- Outputs that may be defined
Note that inputs transform to outputs through the process
functionality/behavior
- Resources consumed that may be defined
Note for the Business Process Framework, resources may be defined
through linkages with the Information Framework
- Activities/Tasks into which the process decomposes that may be defined
These may be embedded in the process description
- A stated or implied value/benefit within the business
May be related to how the process is used within a particular enterprise
- Create, Read, Update, Delete (CRUD) analysis that may be available
This can link with the Information Framework
- Responsible, Accountable, Consulted, Informed (RACI) analysis that may be
available
Typically, this is related to how processes are mapped into organizations
- Key Performance / Quality Indicators (KPIs/KQIs)

These characteristics apply to all levels including extensions and should be included in the definition/description of the process.

A process transforms inputs into outputs. The activities into which it decomposes are performed in some order. The value is created for the customer. The customer may be internal or external.

6.1 Consistency of Inputs and Outputs

The goals, inputs, outputs, and activities of a decomposed process at a lower level are consistent and should match those of the higher-level process, specifically:

- The goals, scope and content of the lower level processes taken together should match those of the higher-level process
- Each input of the higher-level process is the same as an input of a lower level process
- Each output of the higher-level process is the same as an output of a lower level process
- The activities/behavior of the lower level processes taken together should match that of the higher-level process

This ensures the consistency between process extensions to the framework and its further decomposition. “First” and “last” mean the first and last in a sequence of process execution. The Problem Handling example previously described illustrates this.

Customer Problem is input to the L3 process Isolate Customer Problem. It is also input to the first L4 process, Verify Proper Product Use, into which Isolate Customer Problem decomposes. The Root Cause of the Customer Problem is output from the last L4 process, Provide Problem Analysis Completion Notification, it is also output from Isolate Customer Problem. Note that this does not imply that these are the only inputs and outputs of the Isolate Customer Problem process or the processes into which it decomposes. This is particularly true when dealing with higher level processes, such as L2 processes.

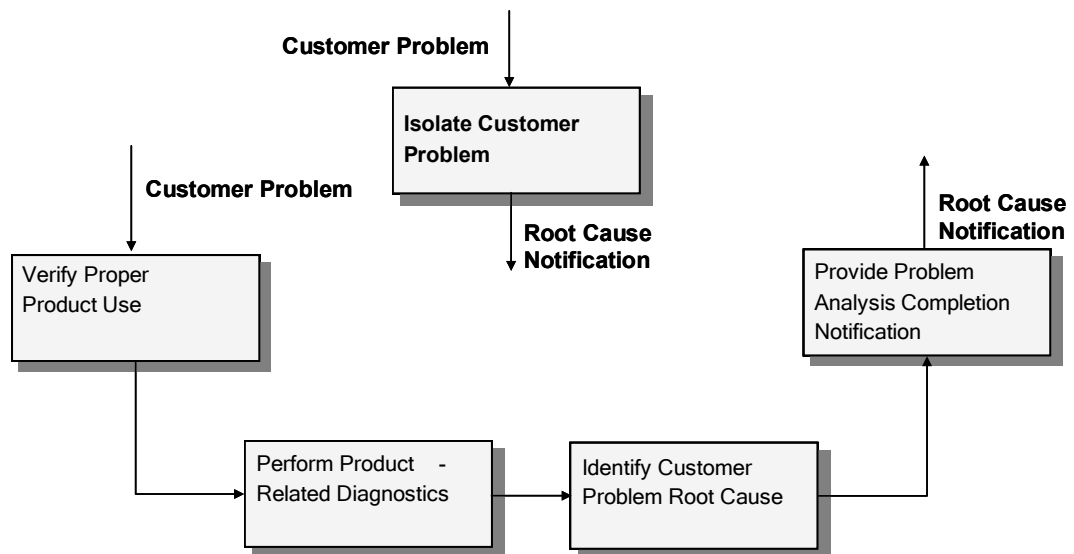


Figure 6-1 - Consistency of Inputs and Outputs

6.2 Additional Decomposition Guidelines

The Business Process Framework represents a non-redundant decomposition; therefore, a process is not duplicated within the framework.

Two different conventions are used for naming Business Process Framework processes. Nouns, often nouns and/or gerunds (a noun form of a verb), such as Fulfilment, Billing and Revenue Management, Customer Problem Handling are used for L1 and L2 processes. This convention indicates a higher-level process that is a combination of tasks (L3). The <verb noun> convention are used for processes that represent more discrete processes, such as tasks (L3, L4, L5, etc). Note that L3, L4, L5 refer to the levels within a process hierarchy and that there may be multiple levels of Business Process Framework process that represent a given process hierarchy level. A bit confusing, but the focus of the Business Process Framework process must be considered when classifying a Business Process Framework process within the levels of a process hierarchy.

Association/translation between process layers, horizontal Business Process Framework processes, is the responsibility of the lower layer. For example, the interaction from Customer Problem Handling to Service Problem Management is expressed in terms a Customer Problem Report. It is the responsibility of Service Problem Management, not Customer Problem Handling, to create the Service Problem Report(s) and associate the Report(s) to the Customer Problem Report. This in part helps ensure that the lifecycle of the entity is managed by a single L2 process and is not split among more than one L2 process.

The process which manages data creation, update, and so forth has a prime responsibility for ensuring that the results of data which it is manipulating via the process are appropriately stored.

When decomposing a process to L3 or to L4, and sometimes below L4, it is helpful to think of the next level of decomposition in terms which represents “life cycle” decomposition. Life cycle decomposition is a technique to use when identifying processes that move an object/entity through its life of interest to the business. L3/L4 processes, each of which represents the task level within the process decomposition, naturally focus on a single state in the life cycle of an object/entity. Typical processes in the decomposition perform “Plan”, “Acquire”, “Use”, “Dispose” tasks. The set of tasks is typically made up of those that plan to use an object/entity, which acquires an object/entity, which uses an object/entity, and which disposes of an entity. Note that life cycle decomposition is just one of the many techniques described here which may be used to identify lower level process elements.

The lifecycle often depends on the category of entity. Plan, Acquire, Use, Dispose processes may take on names specific to the type of entities/objects upon which they act. Shown below are some examples of the names associated with different categories of entities.

Tangible oriented entities, such as products, services, and resources may have the following life cycle related processes:

- Gather/analyze
- Develop
- Deploy
- Assess
- Retire.

Intangible oriented entities, such as strategies and plans may have the following life cycle related processes:

- Research/analyze
- Formulate/prepare
- Approve/commit
- Assess
- Retire.

Activity oriented entities, such as orders and problems may have the following life cycle related processes:

- Issue/Create
- Analyze/diagnose
- Fix, monitor, correct
- Close.

Note that the actual decomposition may only consist of a subset of these types of activities. There are many examples of this type of decomposition in the framework.

Note also that while products and services are intangible entities, they are typically realized by in part some type of physical (tangible) resource. The first set of sample process names represents processes that appear in the Lifecycle Management L1 vertical context; the second set represents processes that appear in the Strategy Management vertical. The third set represents processes associated with activity-oriented entities that span Fulfilment and Assurance.

For example, the decomposition of the L2 Problem Handling includes Create Customer Problem Report (Acquire), Correct & Recover Customer Problem (Use), and Close Customer Problem Report (Dispose) represents processes that appear in the Assurance L1 context vertical. In this example there is no equivalent for the Plan process.

Below L2, beware “/”, “&”, and their variants in process names, which typically means process should be split. Although the Business Process Framework does not always follow the last guideline, processes whose names imply more than one process, such as Implement, Configure, and Activate Service could be split into three processes at the next level of

decomposition. Processes such as these are often combined at a given level of decomposition to manage the number of processes that appear at that level.

6.3 Information Framework Related Guidelines

These guidelines were developed as the Business Process Framework/Information Framework mapping team defined actions taken by processes on SID entities and as the team began to identify L4 processes.

An Aggregate Business Entity (ABE) is a group of closely related (cohesive) entities, such as Customer Problem Report and Service Order. A L2 process manages the life of a small number of ABEs. For example, Service Configuration & Activation manages the life of the Service Order and Service ABEs as defined by the published Business Process Framework/Information Framework L2 to ABE mappings. Customer Problem Handling manages the life cycle of a Customer Problem Report. The L3 processes focus on states within the life cycle of an ABE. For example, Issue Service Order focuses on the creation state; Close Service Order focuses on the completion state. A similar relationship exists for the L3 processes that define Problem Handling. Lower level processes deal with managing the states of the entities associated with the L3 processes and no others.

A process in one domain, such as Customer, Product, Service, does not create instances of entities in another domain. This guideline follows on from the layer's guideline. For example, Problem Handling does not create instances of Service Problem Report. Similarly, Order Handling does not create instances of Service Orders. The two L2 processes reside within the Customer and the Product domains (L1 horizontal process groupings) domains, while Service Problem Report and Service Order are Service domain entities.

6.4 Semantic Analysis

In linguistics, semantic analysis is the process of relating syntactic structures, from the levels of phrases, clauses, sentences and paragraphs to the level of the writing as a whole, to their language-independent meanings, removing features specific to particular linguistic and cultural contexts, to the extent that such a project is possible. The elements of idiom (a fixed distinctive expression whose meaning cannot be deduced from the combined meanings of its actual words) and figurative speech, being cultural, must also be converted into relatively invariant meanings.

Semantic analysis is much like sentence diagramming that we all learned at an early age. The descriptions of L3 processes can be used and analyzed to begin identifying L4 processes.

This is accomplished by performing semantic analysis on the description of a process:

- Look for nouns (entities) upon which actions are performed
- Look for verbs that act on nouns (entities)
- Look for phrases that imply actions on nouns (entities).

Caution: Verbs and phrases may represent processes in other areas of the framework that interact with the processes being identified.

Semantic Analysis should be considered as just another of the many techniques described in this section that can be used to decompose processes. The techniques can be viewed as supplemental to the techniques currently in use within an organization. Often, Semantic Analysis is carried out iteratively with other techniques and may result in updating the description of a process to which Semantic Analysis was originally applied. The goal would be that the final decomposition(s) are consistent with the parent process description(s) from a Semantic Analysis viewpoint.

For simple L3 processes whose description is brief, such as Close Service Performance Degradation Report and Audit Data Collection & Distribution, decomposition to L4 may not be possible. Their descriptions are included here to illustrate this point.

Close Service Performance Degradation Report - The objective of the Close Service Performance Degradation Report processes is to close a service performance degradation report when the service performance has been resolved. These processes monitor the status of all open service performance degradation reports and recognize that a service performance degradation report is ready to be closed when the status is changed to clear.

Audit Data Collection & Distribution - The Audit Data Collection & Distribution processes are responsible for auditing the management information & data collection activities in order to identify possible anomalies such as loss of management information and/or data in the different collection, processing and distribution steps.

Decomposing from L3 Description Example 1 – Isolate Customer Problem

Isolate Customer Problem is the L3 process being decomposed.

The purpose of the **Isolate Customer Problem** processes is to identify the root cause of the customer problem. The responsibilities of these processes include, but are not limited to:

Verifying Configuration; and

Performing specific customer problem diagnostics based on the customer provided information to determine whether the root cause of the customer problem is linked to the underlying services.

The Isolate Customer Problem processes will make the results of the root cause analysis available to other processes. The Isolate Customer Problem processes will update open customer problem report, as required during the assessment, and when the **root cause has been identified**. The Isolate Customer Problem processes will **notify the Track & Manage Customer Problem processes when the analysis is complete**.

The highlighted text represents the candidate L4 processes using the semantic analysis guidelines described previously in this section. Not all of the descriptions resulted in candidate L4 processes. Some were used as part of the description of the L4s or represent the initiation of flows to other processes. For example, the notification to Track & Manage Customer Problem is defined using a flow to that process; updating the open Customer Problem Report is also accomplished by initiating a flow to Track & Manage Customer Problem.

Note that all the nouns (entities) deal with the Customer Problem ABE and its entities, such as Root Cause.

Figure 3.5 - Isolate Customer Problem Decomposition shows the resulting decomposition diagram for Isolate Customer Problem. Some of the process's names represent interpretations that were made based on the analysis of the Isolate Customer Problem description. All of the L4 processes deal with the "isolate" state of the Customer Problem.

The Provide Problem Analysis Completion Notification implies a flow to another process, such as Track & Manage Customer Problem L3 or an L4 process within the Track & Manage decomposition.

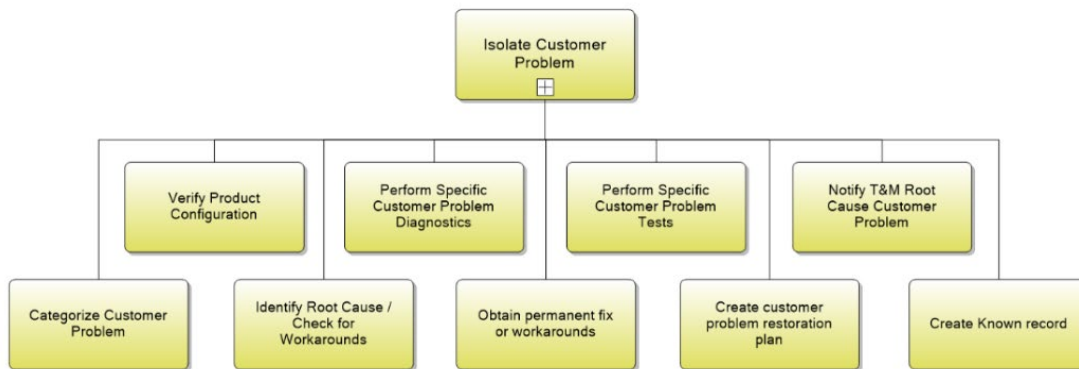


Figure 6-2 - Isolate Customer Problem Decomposition

Below are examples of L4 process descriptions. The descriptions of the L4 processes were based on the description of the L3 process. These L4 processes represent the lowest level to which the current Isolate Customer Problem can be decomposed. Later in this chapter we will discuss how processes can be extended using this same technique of starting with an extended description of a process. Other guidelines for extending the framework will also be provided.

- **Verify Product Configuration**

This process verifies whether the customer is using the purchased product offering correctly.

- **Perform Specific Customer Problem Diagnostics**

This process performs diagnostics based on the customer provided information to determine whether the root cause of the customer problem is linked to the underlying services.

- **Perform Specific Customer Problem Tests**

This process runs tests against specific services.

- **Notify T&M Root Cause Customer Problem**

This process notifies the Track & Manage Customer Problem processes when the analysis is complete.

- **Categorize Customer Problem**

This process is responsible for categorizing the customer problem according to the type of the problem, impact, and standardized criteria at CRM level. Moreover, this activity will be

carried out with appropriate diligence, in order to provide input to the "Track & Manage Customer Problem" process. Analyze customer problems, filter out irrelevant and false problems. Determine the problem severity based on its impact and set the appropriate priority.

- **Identify Root Cause / Check for Workarounds**

Verify the configuration of the customer's product. Look for the root cause of the customer problem. If the cause is determined, recommend a workaround or a temporary fix for the problem.

- **Obtain permanent fix or workarounds**

Check for permanent fixes for the customer and if not available look for possible workarounds as remedy for the problem.

- **Create customer problem restoration plan**

The primary aim is to put a plan to restore the 'normal service' to the customer as quickly as possible. That should include anything that is needed to allow the customer to return to the normal working, in order to minimize business impact

- **Create Known record**

The record includes the details of the problem and the symptoms that occurred, together with precise details of any workaround or fix that can be taken to restore the service and/or resolve the problem.

Other L3/L4 Decomposition Examples

This example illustrates the current L3/L4 process decomposition work being undertaken by the Business Process Framework/Information Framework mapping team. This example illustrates the work accomplished for the L3 processes for the L2 Customer Problem Handling process. Inputs and outputs were derived from the descriptions and implied actions on entities contained in the descriptions of the L3 processes. These, along with semantic analysis, discussed earlier in this chapter, were used to deduce the candidate L4 processes that will make their way into the framework.

The Inputs and outputs can be used to construct/confirm L3 and L4 process flow diagrams. This process shows the work done that resulted in the Problem Handling decompositions.

L3 Process Name	Input Information	Output Information	Candidate L4 Processes
Create Customer Problem Report	I-1) Customer Information	O-1) Customer Problem Report	
	I-2) Customer/ Process Request	O-2) Customer Problem Report	
Isolate Customer Problem	I-1) Customer Problem	O-1) Product Use Verification	Verify Proper Product Use
	I-2) Customer Problem	O-2) Product-Related Diagnostic Results	Perform Product-Related Diagnostics
	I-3) Customer Problem I-4) Problem Related Diagnostic Results	O-3) Customer Problem Root Cause	Identify Customer Problem Root Cause
	I-5) Customer Problem I-6) Customer Problem Root Cause	O-4) Customer Problem Analysis Completion Notification	Provide Customer Problem Analysis Completion Notification
Report Customer Problem	I-1) status of CustomerProblemReport	O-1) notifications of any changes and management reports	

Figure 6-3 – Customer Problem Handling L4 Processes

Figure 6-4 Resource Provisioning L4 Processes shows similar work accomplished for Resource Provisioning L3 processes.

L3 Process Name	Input Information	Output Information	Candidate L4 Processes
Allocate & Install Resource	I-1) Service Order I-2) LogicalResource	O-1) ResourceSpecification	Check Resource Feasibility
	I-3) Resource Order	O-2) ResourceSpecification O-3) ResourceConfig	Allocate Resource
	I-4) ResourceSpecification	Nil	Install Resource
	I-5) Supplier/Partner	O-4) ResourceSpecification	Check Resource Availability
Configure & Activate Resource	I-1) Resource Order	O-1) ResourceOrder	Plan Resource Configuration and Activation
	I-2) ResourceConfig I-3) Resource	O-2) ResourceSpecification	Configure Resource
	I-4) ResourceSpecification I-5) ResourceConfig	O-3) ResourceAlarm	Notify Potential Resource Trouble
	I-6) ResourceSpecification I-7) ResourceUsage I-8) ResourceConfig	O-4) AdministorResourceData O-5) Resource	Update Resource Inventory

Figure 6-4 - Resource Provisioning L4 Processes.

Figure 6-5 Resource Trouble Management L4 Processes shows the decomposition for Resource Trouble Management L3 processes.

L3 Process Name	Input Information	Output Information	Candidate L4 Processes
Create Resource Trouble	I-1) Resource Alarm Event Record	O-1) Resource Trouble	Generate Resource Trouble
	I-2) Service Trouble I-2) Resource Order	O-2) Resource Trouble	Convert Report to Resource Trouble
	I-3) Resource Trouble I-4) Resource Restoration Time	O-2) Resource Trouble	Estimate Time for Restoring Resource
Localize Resource Trouble	I-1) Resource Trouble I-2) Current Resource Configuration I-3) Resource Facing Service Configuration	O-1) Resource Trouble O-2) Resource Configuration Verification	Verify Resource Configuration
	I-4) Resource Trouble I-5) Resource Configuration Verification I-6) Resource Trouble Diagnostics Specifications I-7) Resource Trouble Diagnostics Methodology Template(New)	O-3) Resource Trouble O-4) Resource Trouble Root Cause Analysis	Perform Specific Resource Trouble Diagnostics
	I-8) Resource Trouble I-9) Resource Trouble Test Specifications I-10) Resource Trouble Root Cause Analysis	O-5) Resource Trouble O-6) Resource Trouble Test Results (Created) O-7) Resource Trouble Root Cause Analysis	Perform Specific Resource Trouble Tests

Figure 6-5 - 13 Resource Trouble Management L4 Processes

7. Audit Checklist

This section contains a checklist of the Business Process Framework principles for its use and application within an organization.

Principle	
Business Process Framework - 01	The Business Process Framework is an activity-based process decomposition model
Business Process Framework - 02	The Business Process Framework Process Elements are comprehensive for a Service Provider
Business Process Framework - 03	A Business Process Framework Process Element is a succession of activities with a control flow
Business Process Framework - 04	(Best Practice): A Business Process Framework Process Element has a goal, value proposition, inputs, outputs. It consists of activities and uses resources. It has a CRUD and RACI model.
Business Process Framework - 05	The Business Process Framework is decomposed from notional Level 0 to more granular levels – Levels 1, 2 and 3 (and some of level 4) are addressed so far. An agreed endpoint (i.e., a level below which decomposition does not proceed) is not yet defined.
Business Process Framework - 06	It is not the purpose of the Business Process Framework to address the detailed processes and procedures of an enterprise
Business Process Framework - 07	The Enterprise Domain is generally decomposed to Level 3 only (but specific areas that represent particular priorities have been decomposed further).
Business Process Framework - 08	The goals, inputs, outputs, and activities of decomposed Process Elements at a lower level are consistent with the higher-level Process Element. In particular, the input of the first lower level Process Element is the same as the input of the higher level Process Element; the output from the last lower level Process Element is the same as the output of the higher level Process Element; the goals of the lower level Process Elements taken together should match the goal of the higher level Process Element; the activities of the lower level Process Elements taken together should match the activity of the higher level Process Element.

Principle	
Business Process Framework - 09	(Best Practice): each Business Process Framework Process Element should identify its associated Information Framework ABEs
Business Process Framework - 10	A particular Process Element will occur only once in the Business Process Framework; there is no replication
Business Process Framework - 11	Process flows in the Business Process Framework are control flows, defining the sequence order in which activities are performed.
Business Process Framework - 12	The swim lanes in a process flow are consistent within themselves and with respect to lower level decompositions.
Business Process Framework - 13	The Business Process Framework models success scenarios. Error conditions are not in scope.
Business Process Framework - 14	Dynamic aspects of process modelling are outside the scope of the Business Process Framework.
Business Process Framework - 15	Terminology and naming conventions are <Noun> for Level 1 & 2 Process Names, <Verb Noun> of Level 3 Process Names, <Noun Verb> for events.
Business Process Framework - 16	Responsibility for association / translation between layers is generally positioned at the lower layer.
Business Process Framework - 17	A process has prime responsibility for ensuring that the results of data manipulation are sowed appropriately.

8. Administrative Appendix

9.1 Acknowledgements

This release of the Business Process Framework is the result of the combined efforts of a large group of individuals from companies all over the world. Most noteworthy is the participation of numerous service providers. The knowledge and commitment in providing contributions and participating in discussions are greatly appreciated. Contributors over the program leading to previous Business Process Framework/eTOM releases were acknowledged in those documents

The team looks forward to continued input and involvement for ongoing work on the Business Process Framework. Thank you for making this the acknowledged, best framework for Telecom and Information Services business processes.

See main document (GB991 Concepts and Principles) for other acknowledgements.

9.2 Document History

9.2.1 Version History

Version Number	Date Modified	Modified by:	Description of changes
0.21	November 2006	Philip Williams	Document launch
1.0	December 2006	Mike Kelly	Formatting for first issue of document
1.1.	February 2007	Tina O’Sullivan	Updates from AC
1.2	June 2009	Alicja Kawecki	Minor updates to reflect TM Forum Approved status
1.2	June 2009	Alicja Kawecki	Minor updates to reflect TM Forum Approved status
1.3	Jan 2010	Mike Kelly (with some updates by Ken Dilbeck)	Small terminology changes to use “Business Process Framework” and to update diagrams, Also, small textual clarifications; incorporation of member review comments
1.4	March 2010	Alicja Kawecki	Minor cosmetic updates for web posting
1.5	April 2011	Alicja Kawecki	Updated to reflect TM Forum Approved status
1.6	September 2012	John Reilly	Added decomposition guidelines chapter
1.7	October 2012	Alicja Kawecki	Corrected notice, minor cosmetic corrections prior to web posting and Member Evaluation

Version Number	Date Modified	Modified by:	Description of changes
1.8	July 2013	Alicja Kawecki	Updated to reflect TM Forum Approved status
1.8.1	November 2014	Alicja Kawecki	Applied rebranding, corrected Notice
1.8.2	March 2015	Alicja Kawecki	Updated cover, header, footer and Notice to reflect TM Forum Approved status
15.0.0	May 2015	Avi Talmor	Notes on domain addition
15.0.1	May 2015	Alicja Kawecki	Updated cover, Notice
15.0.2	16 December 2015	Alicja Kawecki	Updated cover, header and Notice to reflect TM Forum Approved status
16.0.0	05-Jul-2018	Alfred Anaya-Dubernard	Updates to align with Fx18 and GB991
16.0.1	06-Jul-2018	Adrienne Walcott	Formatting/style edits prior to R18 publishing
16.0.2	24-Sep-2018	Adrienne Walcott	Updated to reflect TM Forum Approved Status
19.0	23-Jul-2019	Alfred Anaya-Dubernard	Updates to align with Fx19 and GB991
19.0.1	17-Oct-2019	Adrienne Walcott	Updated to reflect TM Forum Approved Status
22.5	07-Dec-2022	Alfred Anaya-Dubernard	Updated to reflect changes to newest version 21.5

9.2.2 Release History

Release Number	Date Modified	Modified by:	Description of changes
Release 7.0	December 2006	Mike Kelly	Formatting for first issue of document
8.1	Jan 2010	Mike Kelly (with some updates by Ken Dilbeck)	Small terminology changes to use “Business Process Framework” and to update diagrams; Also, small textual clarifications; incorporation of member review comments
12.0	September 2012	John Reilly	Added decomposition guidelines chapter
14.5.0	November 2014	Avi Talmor	Added note regarding change to core processes decomposition
15.0	May 2015	Avi Talmor	Notes on domain addition
18.0.0	June 2018	Alfred Anaya-Dubernard	Updates to align with Fx18 and GB991
18.0.1	24-Sep-2018	Adrienne Walcott	Updated to reflect TM Forum Approved Status
19.0.0	23-Jul-2019	Alfred Anaya-Dubernard	Updates to align with Fx19 and GB991

Release Number	Date Modified	Modified by:	Description of changes
19.0.1	17-Oct-2019	Adrienne Walcott	Updated to reflect TM Forum Approved Status
Production	13-Feb-2023	Adrienne Walcott	Updated to reflect TM Forum Approved Status